GAUGE, NOT LEVER: DISSOCIATING READABLE CORRECTNESS FROM ITS CAUSAL REPAIR CHANNEL IN A 27B REASONING MODEL

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ABSTRACT

Large language models linearly encode whether they are about to answer correctly (probe AUC up to 0.94), and recent work shows this readout resists steering — but why, and what *does* causally control correctness? On depth-controlled ontology-reasoning tasks with Gemma 3 27B, we first show the readout is not necessary: deleting everything linearly readable about correctness — the full rank-9 INLP subspace at all five readable layers, every token — costs nothing ($+0.05$ $[-0.07, +0.20]$), while matched-rank random subspaces are catastrophic (-0.38). Correctness is instead controlled through a compact channel at concept-mention tokens: a rank-8 basis repairs held-out failures ($+0.25$), passing random-subspace and matched-noise specificity controls. A natural correct-minus-incorrect class-mean vector is null at full dimension yet repairs strongly ($+0.34$) when concentrated into that channel; shuffled-label and sign-flip controls show the labels carry the effect, and a fully answer-free variant reaches $+0.40$ on a 0.12 baseline where majority-vote self-consistency scores 0.000. The channel is causally potent yet not decodably privileged (probe accuracy inside the random-subspace null) — the inverse of the field's decodable-but-not-causal refrain — and removal experiments show its site demands the state's content wholesale: the lever is write-compact, but what it writes into is carried broadly. Positive results are one model and task family, with the raw-axis non-necessity replicating in a second model; every causal experiment was pre-registered with branch-complete decision rules (23 verdicts, 7 confirmed directional predictions).

1 INTRODUCTION

A linear probe on a language model's residual stream can tell you, before generation begins, whether the answer about to be produced will be correct. This is by now a replicated fact (arXiv:2605.05715, 2026; arXiv:2604.13068, 2026; arXiv:2604.05655, 2026). It is also, on its face, an invitation: if the model internally represents "I am about to get this wrong," surely writing the opposite into the residual stream should help. We spent months accepting that invitation — raw-direction steering, optimized vectors, low-rank interchange over the probe subspace, decode-time gated injection — and repaired nothing. Independently and concurrently, three groups hit the same wall: the failure signal is decodable but steering it is null (arXiv:2605.05715, 2026; arXiv:2604.13068, 2026), a phenomenon one group names *epiphenomenal correctness* (arXiv:2605.23315, 2026). The field's diagnosis stops at the null. This paper asks the question the other way around: if the readable signal is not how correctness is controlled, what is — and what, precisely, is the readable signal *for*?

We work on abductive ontology reasoning (InAbHyD), where task difficulty is controlled by proof depth and a strict string-level scoring of the proposed hypothesis (*strong* correctness) gives a binary outcome to predict and intervene on. The statistical unit throughout is the task instance (row): every effect is a pooled difference in $P(\text{strong})$ against an in-job baseline with row-cluster bootstrap confidence intervals, and every causal experiment was pre-registered with branch-complete decision rules before unblinding. Our answer, compressed to a sentence: *in a model that visibly predicts its own failures, the readable correctness signal and the causal repair channel are different objects — the entire readable subspace can be deleted without behavioral cost, while repair flows exclusively* *through a low-rank channel that carries the label-specific content of natural success yet is itself no more decodable than chance — gauge and lever, anatomically separated.*

Concretely, we contribute: **(i)** a readable-stack erasure result with destructive matched-rank controls and a stated variance mechanism: nothing linearly readable about correctness is necessary (Section 5.2); **(ii)** a localized, compact causal channel — rank 8 at concept-mention tokens — that survives fresh-row and specificity guards (Sections 5.3–5.4); **(iii)** a necessity dissociation: the same natural class-mean vector is behaviorally null at full dimension and strongly reparative concentrated into the channel — and, in the removal direction, content-necessity at the channel's site that is broad rather than thin (Section 5.5); **(iv)** a label-specificity battery (shuffled labels, sign-flips, matched noise, random subspaces) that no steering paper in the verified corpus runs, falsifying both external reviewers' predictions; **(v)** causal control without decodable privilege — the inverse dissociation of the readable-but-inert readout; **(vi)** fully answer-free repair with a measured deployment profile: $+0.40$ where self-consistency scores 0.000, replicated on a disjoint fresh draw ($+0.28$), beneficial when misfired on already-correct rows — and with the addressing gap reduced to a protocol question, since the gauge itself selects the correct write address among steered candidate branches; and **(vii)** a pre-registration methodology — branch-complete decision rules before unblinding across 23 registered verdicts, seven confirmed directional predictions, and token-level determinism gates — that we propose as a working standard for intervention claims in interpretability.

2 Dataset and task

We generate InAbHyD abductive-reasoning instances: synthetic ontologies (taxonomic trees over invented concepts with inherited properties) in which the model must propose the hypothesis that best explains a set of observations. Tree height (1–4) controls proof depth; we run two task framings and use *property* (which property does the target concept have?) as primary, with *subtype* and a second model as scope checks. Each model answers 44,000 instances (1k/2k/3k/5k per height and task), scored two ways: *strong* correctness requires the exact gold hypothesis after canonicalization; *weak* accepts entailed variants. All causal work predicts and intervenes on strong correctness. Intervention experiments draw balanced height-3/4 rows from this pool under a strict provenance discipline: development rows (13), fresh-row guard sets (26+6), a 96-row natural-state capture set, and collateral sets of naturally-correct rows are pairwise disjoint, seeded, and frozen before use, so no basis, vector, or amplitude is ever fit on a row it is tested on (Appendix A).

3 Methods

Probes and the readable subspace. Correctness probes are class-balanced logistic regressions on residual-stream activations at the final prompt token, trained on held-out splits of the stage-1 dataset at five layers (L15–L53 in Gemma 3 27B). The *readable subspace* is constructed by iterative nullspace projection: at each layer we run INLP until probe AUC collapses to chance, orthonormalize the accumulated directions across all five layers, and obtain a rank-9 stack — everything any linear probe in our battery can read.

The erasure family. Erasure is a mean-projection clamp: at every token position, through prompt processing and decoding, the projection of the residual onto each stack direction is clamped to that direction's global scalar train-projection mean. Controls apply the *identical estimator and clamp* to matched-rank random stacks. Registered telemetry records within-forward-pass projection variance per direction (the mechanism check), and every arm reports $P(\text{strong} \mid \text{parsed})$ alongside $P(\text{strong})$ to separate content damage from format damage.

Interchange on the focus state. For a failing row, a *hint-delta* is the difference between residual states under a concept-hinted prompt and the unhinted prompt, computed on the longest common token block and restricted to the gold concept's mention positions. Repair arms add rank-k PCA reconstructions of these deltas at layer 30; bases are always fit leave-one-row-out (or on disjoint rows entirely), and additions are norm-matched per position to the reconstruction norms. Controls substitute random orthonormal subspaces, matched-norm Gaussian noise, random positions, or the delta mean alone at identical norms.

Control	Mundane account it kills	Result
Random positions	any location works	null
Matched-norm noise	any perturbation this size helps	destructive
Random subspaces (norm/rank-matched)	any k directions work	null
Shuffled-label class-means	any difference-shaped vector works	null
Sign-flipped vector	geometry, not content	harmful
Shuffled-ridge coefficients	predicted content is generic	informative fail
Random stacks, identical estimator	erasure procedure is toothless	catastrophic
Dose ladders ($\times\frac{1}{2}, \times 1, \times 2$)	more amplitude, more repair	monotone damage
Coordinate-permuted basis	spectrum-matched removal breaks	near-free
Typical-state (mean) replacement	the break is deleted energy	floors
Keep-only-basis complement deletion	the 8 dims carry the site	floors

Table 1: **Every positive result is bracketed by the control that would explain it away.** Sampling baselines (self-consistency, best-of-n) and collateral slices on naturally-correct rows complete the battery.

The class-mean family. Natural-state arms replace hints with observation: we capture unhinted concept-position states on 96 provenance-disjoint rows (balanced correct/incorrect by strong label), take per-row position-means, and form the class-mean $\bar{h}_{\text{correct}} - \bar{h}_{\text{incorrect}}$. Variants apply it raw or projected into the rank-8 basis; the *donor-free* variant fixes amplitude to a single pooled scalar so no per-instance information enters. Label controls recompute the mean under shuffled labels (identical geometry, norm, positions) or flip its sign; removal arms project states onto the orthogonal complement of a frozen basis, replace them with the dataset-typical mean state, or retain only the basis component.

Control taxonomy. Table 1 is the design's spine: each control is matched to the mundane account it eliminates.

Protocol. Every causal experiment was pre-registered with branch-complete decision rules — a named branch for every reachable outcome pattern, wording included — committed before unblinding; seven directional predictions were registered and all seven confirmed. Generation is seed-deterministic per (row, arm): replication-gate arms reproduce prior jobs token-for-token ($416/416$, $736/736$, and $456/456$ generations across the necessity and addressing sequences alone), which converts job-stitching and cross-experiment anchoring from an assumption into a checked gate. Statistics: row-cluster bootstrap (10k draws, fixed seed, percentile CIs) against in-job baselines; claim-bearing contrasts near zero additionally get BCa and leave-one-row-out checks; every null is reported with its minimum detectable effect.

4 Related work

Readable but unsteerable. Our closest priors found the failure regime decodable (71.6%) with 29 fixed-linear-steering configurations null, concluding *entanglement* from a destructive LEACE erasure (arXiv:2605.05715, 2026), and detection-without-correction across seven model families (arXiv:2604.13068, 2026); arXiv:2605.23315 (2026) coins "epiphenomenal correctness" from cross-model convergence with small ablation effects. All three stop where our contribution starts: none localizes the causal variable or demonstrates repair, and our harmless full-stack erasure with destructive matched-rank controls adjudicates the entanglement account directly (Section 5.2). Correlational correctness probing supplies the readable half as background (arXiv:2604.05655, 2026; arXiv:2504.05419, 2025; arXiv:2602.06022, 2026).

Steering reasoning with difference vectors. The recipe family exists and we did not invent it: difference-in-means and related vectors can raise reasoning accuracy (arXiv:2601.19847, 2026; arXiv:2509.18116, 2025; arXiv:2505.12189, 2025), trained steering vectors recover most of RL's reasoning gains (arXiv:2509.06608, 2025), and probe-gated deployment is published (arXiv:2601.19847, 2026). None of this work shows necessity, label-specificity, or decodability dissociations; SAE-RSV (arXiv:2509.23799, 2025) shows raw vectors are mostly noise and filtering helps — we show a causally-identified filter is, in our setting, necessary, and what it retains is label-specific.

The methods bar. We adopt the state-encoding explicitness demanded by causal-abstraction critiques (arXiv:2507.08802, 2025) (fixed linear PCA bases, no trained alignment maps), the harmless-versus-pernicious vocabulary of arXiv:2511.04638 (2026) for the probe readout, mean-projection over INLP for the erasure operation (arXiv:2506.11673, 2025), and the subspace-illusion discipline of Makelov et al. (2023) (with Wu et al. (2024) as counterpoint) via fresh-row and basis-provenance guards. Steering-rigor critiques (arXiv:2501.17148, 2025; arXiv:2407.12404, 2024; arXiv:2410.17245, 2024; arXiv:2507.11771, 2025; arXiv:2411.07213, 2024) function here as the checklist our controls answer (Table 1).

Concurrent convergence. A concurrent frontier-lab result reports the same structural phenomenon on a different variable and model family: a small privileged subspace is causally dominant while decodable-but-inert content lives elsewhere (Anthropic Interpretability, 2026), independently replicated on a relative of our second model. We cite this as convergence on the anatomy — not prior art on correctness, repair, or the label battery.

5 Results

All effects are pooled differences in $P(\text{strong-correct})$ against in-job unhinted baselines, with row-cluster bootstrap 95% confidence intervals (10,000 draws, fixed seed); rows, not generations, are the unit of analysis. Every causal experiment below was pre-registered with branch-complete decision rules before unblinding, and replication-gate arms reproduce prior jobs token-for-token (Section 3).

5.1 The gauge: correctness is linearly readable

A linear probe on residual-stream activations at the final prompt token predicts whether the model's free-form answer will be strong-correct, before generation begins: test AUC rises from 0.786 at L15 to 0.902 at L53 in Gemma 3 27B, and reaches 0.940 in Qwen3.5-27B. The readout survives entity-name scrambling and beats every metadata baseline (height, prompt length, class balance). This much is consistent with a now-replicated literature: correctness and error are decodable pre-generation across model families, and the readable signal resists steering (arXiv:2605.05715, 2026; arXiv:2604.13068, 2026; arXiv:2605.23315, 2026). The question the rest of this section answers is what that readability is worth causally.

5.2 The gauge is not necessary

Design. If the readable correctness signal participates in producing correct answers, deleting it should hurt. We deleted it completely. From the five probe directions we constructed the full rank-9 INLP-reachable correctness subspace — every direction any linear probe in our battery can read, at all five readable layers — and applied a mean-projection clamp at *every* token position, through both prompt processing and decoding. As controls, we applied the *identical* estimator and clamp to matched-rank random stacks. Sixteen balanced rows (8 naturally correct, 8 incorrect), six conditions, k=8 sampled generations each.

Result. Erasing the entire readable stack costs nothing detectable: $+0.047$ $[-0.070, +0.203]$ against the in-job baseline, with the single-axis continuity gate reproducing the historical null (-0.016 $[-0.094, +0.070]$). The identical operation on matched-rank random stacks is catastrophic: -0.380 $[-0.586, -0.182]$ pooled across three draws, with correctness among still-parseable outputs collapsing from 0.444 to ~ 0.01 — content demolition, not formatting damage. The paired contrast (readable stack $-$ random family) is $+0.427$ $[+0.214, +0.643]$.

Mechanism, stated rather than hidden. The registered variance telemetry explains *why* the readable stack erases so cheaply: it carries 10–1300× less within-forward-pass projection variance than the random stacks — the clamp removes almost nothing where the readout lives. We therefore claim **non-necessity** of the readable subspace, never that the axis is "specially inert" among same-norm

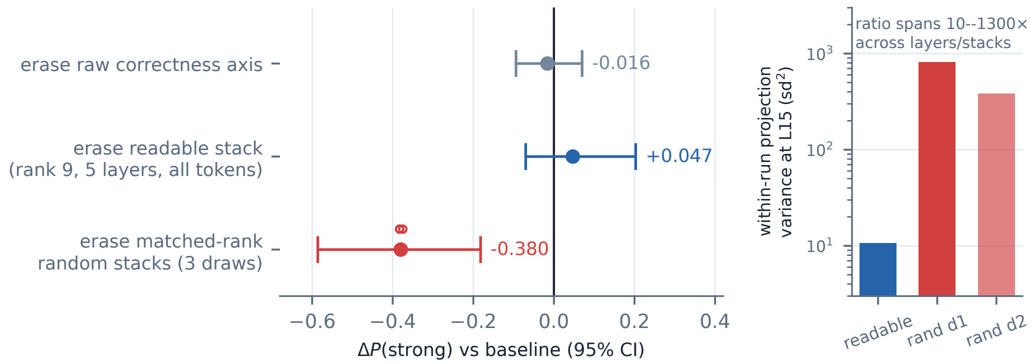

Figure 1: **Deleting everything readable costs nothing; deleting matched-rank random subspaces is catastrophic.** Pooled ΔP(strong) with row-cluster bootstrap 95% CIs; 16 balanced rows, k=8. Inset: within-run projection variance per stack (log scale) — the readable stack carries 10–1300$\times$ less than the random stacks, the registered mechanism for cheap erasure.

directions: the destructive controls prove the erasure family has behavioral teeth at matched rank and norm, and the low-variance mechanism explains the asymmetry. By the same discipline, the null is bounded, not proven zero: the minimum detectable effect at n=16 rows is ≈ 0.13.

What this adjudicates. The nearest rival account holds that readable correctness is *entangled* with task computation — their LEACE-style erasure damaged accuracy by 3.6pp, and they conclude the signal cannot be removed cleanly (arXiv:2605.05715, 2026). Our strictly larger erasure is free while matched-rank controls destroy, which excludes the destruction-on-removal form of entanglement in our setting. The weak, redundant-carrier form survives, and we say so: retrained probes still decode correctness after the scrub (information is redundant), yet no readable axis is load-bearing. Readability here is a gauge on the dashboard — which is precisely why months of steering it, by us (arXiv:2604.13068, 2026; arXiv:2605.23315, 2026, replicating), repaired nothing.

5.3 The lever exists and is localized

The failure is a proposal failure. Behaviorally, the model's errors are not ignorance: given the gold hypothesis among candidates, accuracy jumps from 0.036 to 0.839; hinted with the target concept, to 0.955–1.000; yet the gold hypothesis is absent from sixteen sampled proposals on roughly two-thirds of failing rows. The model recognizes the answer it cannot propose — a *recognition gap* localized to target-concept focus.

Patching the focus state repairs generation. Replacing residual states at the gold concept's mention positions (layer 30) with their hint-conditioned counterparts repairs free-form generation on fresh rows disjoint from all development: $+0.255$ $[+0.111, +0.413]$ (the development-row anchor, reported as a labeled selected-row figure, is $+0.491$ $[+0.28, +0.71]$). Potency is positional: per-token, concept-mention patches are $\sim$6$\times$ more effective than matched random positions, whose family is null. The token route is ruled out: masking the hint span costs nothing (natural decode attention to it is 0.5%), literal hint-token KV transplants are insufficient with attention verifiably flowing, and the masking$\times$reversion combination survives at 0.981 — the carrier is the unpatched layers' states at concept positions, not the hint tokens.

5.4 The lever is compact and specific

A rank-8 leave-one-row-out PCA basis of the hint-deltas carries the repair to rows that contributed to neither the basis nor the row selection: $+0.231$ $[+0.111, +0.365]$, i.e. 91% of the in-job patch effect (rank-4 under-transfers at 57% — rank 8 is the portable core). The pre-registered specificity ladder then separates the channel from its footprint: the rank-8 reconstruction repairs $+0.245$ $[+0.111, +0.394]$; its mean alone recovers 35%; random subspaces matched in mean, norm, rank, and position do nothing (-0.041); matched-norm noise at the same positions is destructive (-0.088 $[-0.180, -0.017]$); paired rank-8$-$noise is $+0.333$ $[+0.186, +0.489]$. The specific directions carry the repair. Geometrically, the core is disjoint from the gauge: its projection onto the readable stack is 0.0002, *below* the random-subspace null (0.0017), and while sparse-dictionary decoders see it, they see it redundantly — dictionary-visible, not sparse-small.

5.5 THE NECESSITY DISSOCIATION: WHAT THE CHANNEL CARRIES

The lever of Sections 5.3–5.4 was built from hint-derived states. This section removes the hints, then the donors, then — in the final experiments — the assumption that the channel is where success itself lives. Throughout, the vector under test is the *natural class-mean*: the mean difference between states captured (without hints, at the same concept-mention positions, layer 30) on rows the model spontaneously gets right versus wrong — 96 capture rows, provenance-disjoint from every test row.

The same vector, five fates. At matched total norm on 26 held-out failing rows: (i) applied *full-dimensional*, the class-mean is null: $+0.043$ $[-0.120, +0.207]$; (ii) *concentrated into the rank-8 channel* by projection, it repairs strongly: $+0.341$ $[+0.202, +0.495]$ (BCa-stable; leave-one-row-out point range $[+0.315, +0.355]$); (iii) with *labels shuffled* before the mean is taken — identical geometry, norm, and positions — it does nothing: -0.043 $[-0.103, +0.006]$ pooled over four draws, paired real-vs-shuffled $+0.385$ $[+0.245, +0.534]$; (iv) with its *sign flipped*, it is actively harmful: -0.120, CI < 0; (v) made *fully donor-free* — direction from natural outcomes only, amplitude a single pooled scalar, zero per-instance answer information — it produces the strongest repair in the program: $+0.399$ $[+0.245, +0.558]$. Both external reviewers predicted the shuffled arm would match the real vector; it did not. The labels carry the effect.

Against baselines nobody reports. On these hardest rows (baseline 0.120), majority-vote self-consistency (Wang et al., 2022) over eight samples scores 0.000 — errors are correlated across rollouts, so resampling regenerates the same wrong commitment — and a best-of-8 oracle ceiling reaches only 0.192, under half the steered result. Per-row: 16 improved, 10 unchanged, 0 worsened. Where prompting-side compute is the standard of comparison for steering claims (arXiv:2501.17148, 2025), this is the regime in which a causal write reaches states that sampling provably cannot.

The inversion. Is the channel privileged *readably*? A probe restricted to the rank-8 subspace decodes natural correct-vs-incorrect at 0.70–0.72 AUC — inside the random-subspace null (p95 $= 0.721$; full-dimensional ceiling 0.807). The subspace that exclusively carries the repair holds no privileged linearly-decodable information about natural outcomes. This is the mirror image of the field's decodable-but-not-causal refrain (arXiv:2605.05715, 2026; arXiv:2605.23315, 2026), demonstrated within one experiment pair: the readable thing is deletable (Section 5.2); the causal thing is not readable. It also sharpens a live methods debate: raw difference-in-means vectors are mostly noise and filtering helps (arXiv:2509.23799, 2025) — here, causal filtering is *necessary* (raw null, projected potent), and what the filter retains is label-specific content.

Deployment profile, measured. Firing the donor-free vector on rows the model already gets right *helps*: $+0.266$ $[+0.086, +0.469]$ ($0.727 \rightarrow 0.992$) — false-positive firings with correct addressing are beneficial, so gauge-gating is optional rather than load-bearing. The one answer-adjacent ingredient is *addressing*: written at all taxonomy-concept positions indiscriminately, the effect cancels ($+0.029$, n.s.), and every pre-named answer-free position policy fails. The mechanism readout explains why: the vector acts as a positional commitment *command*, pulling proposals toward whichever concept is marked ($\sim 50\%$ regardless of correctness) — content is the verb; the marked concept is the object. Two registered follow-ups then close most of the remaining gap. The repair replicates on a second, fully disjoint draw of failing rows under its original protocol ($+0.279$ $[+0.182, +0.377]$ on 57 rows; 62% of the anchor), while a frozen-basis, pooled-amplitude variant of the same vector transfers to correct rows but does not repair failures (-0.004) — so in-job leave-one-out basis fitting and per-row norm matching are part of the recipe, and we say so. And the addressing loop itself works at the selection stage: enumerating candidate concepts from the prompt, firing the lever at each candidate's positions in parallel branches, and letting the *gauge* score the steered prefills selects the gold-equivalent branch essentially always (gauge-selected =

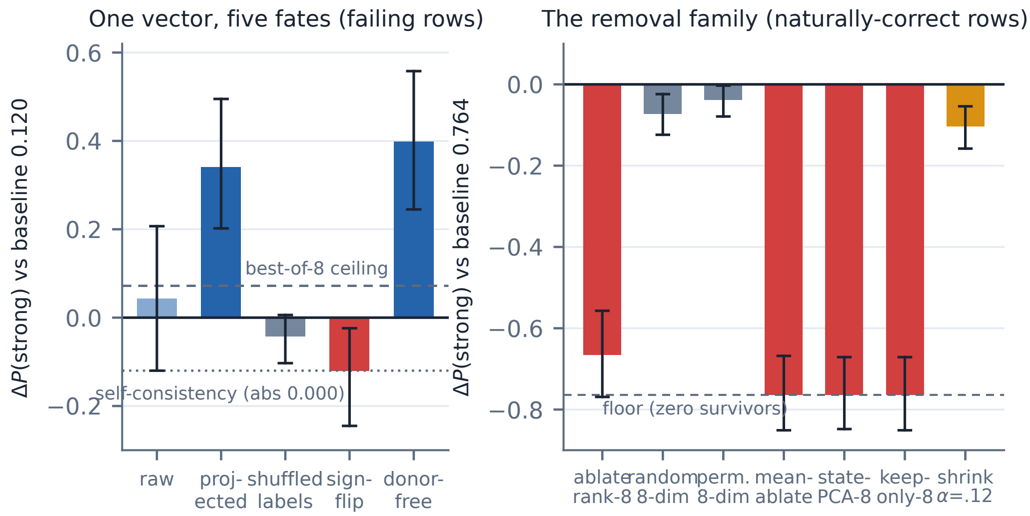

Figure 2: **One vector, five fates — and the removal side.** Left: the natural class-mean at matched norm is null full-dimensional, potent concentrated into the channel, inert with shuffled labels, harmful sign-flipped, and strongest fully donor-free ($+0.399$ on a 0.120 baseline; self-consistency 0.000, best-of-8 0.192). Right: on naturally-correct rows, every content-destroying removal at the lever's site floors success regardless of target subspace, while content-preserving perturbations are an order cheaper. All arms pre-registered; row-cluster bootstrap 95% CIs.

oracle to three decimals; registered selection-signal gate $+31.8$ $[+23.8, +40.1]$). The one full composition run to date paired that working selector with the non-transferring write variant and repaired nothing — an instructive negative: what still separates us from a closed answer-free repair loop is protocol transfer at the write, not answer-adjacent knowledge in the addressing.

The necessity coda: what removal reveals. Sufficiency established, we ran the removal direction on 46 fresh naturally-correct rows (baseline 0.764; all gates verbatim against prior jobs, $416/416$ and $736/736$ generations token-identical). Projecting out the frozen rank-8 basis at the gold concept positions abolishes natural success (-0.666 $[-0.769, -0.557]$) while matched-rank random subspaces (-0.072) and a coordinate-permuted, spectrum-matched basis (-0.038) cost almost nothing. Registered telemetry, however, shows the fitted basis carries 97.7% of the state norm at those positions, so a second registered experiment matched the *energy* rather than the rank: replacing the states with the dataset-typical mean — preserving norm and the giant-magnitude dimensions, destroying row-specific content — floors success entirely (-0.764; zero surviving generations), as does removing the energy-optimal top-8 state-PCA subspace, and, decisively, as does *keeping only* the 8 repair dimensions while deleting their complement. A proportional 12% shrink at the same perturbation norm costs only -0.103, and the sign-flipped class-mean at matched norm beats matched-norm noise (-0.696 vs. -0.351; paired -0.345 $[-0.455, -0.243]$ — a seventh pre-registered directional prediction confirmed). The joint reading is scoped and, we believe, more interesting than the thin-channel necessity it replaces: at the lever's site, natural success requires the state's specific content *wholesale* — **the lever is write-compact, but what it writes into is carried broadly**.

5.6 Scope and cross-model replication

Positive lever claims are Gemma-3-27B, property task. The scope checks: **Subtype** replicates the behavioral story (recognition gap $33/48$; candidates 1.000) but residual-state repair is only suggestive — a weaker, row-sparse analogue at a different depth ($+0.117$ $[+0.008, +0.254]$ at L35, carried by $3/13$ rows). **Qwen3.5-27B** replicates raw-axis non-necessity ($+0.070$ $[-0.031, +0.188]$; its controls are also non-destructive, so this is support, not a replication of the destructive-control profile) — and then the port becomes informative in its own right. The same recognition-gap failure mode exists (hint lift $+0.523$); a concept-position carrier exists at relative depth 0.67 vs.

	Gemma / property	Gemma / subtype	Qwen / property
Gauge readable	✓ 0.90	✓	✓ 0.94
Readout-axis non-necessity	✓	✓	✓ (controls also null)
Full-stack non-necessity	✓	—	—
Lever repair + specificity	✓ rank 8	suggestive	✓ motif, $k^*\approx 16$
F-series dissociations	✓	—	✓ inverted (raw-potent)
Content necessity at lever site	✓ broad, not thin	—	—

Table 2: **Scope matrix.** Checks are pre-registered pooled results; "—" means untested, not negative.

Gemma's 0.48 (concept-replace $+0.175$ $[+0.100, +0.258]$); the channel rank is $\sim$16 vs. 8, with recovery climbing $48\% \rightarrow 95\%$ of the carrier across ranks $8 \rightarrow 64$ while random 64-dim subspaces at matched norm sit at baseline. The content result *inverts*: in Qwen the *raw* class-mean repairs ($+0.120$ $[+0.016, +0.224]$; real-vs-shuffled paired $+0.182$ $[+0.070, +0.299]$; sign-flip harmful, a fifth confirmed registered prediction) and projection into the compact channel *dilutes* it — while no outcome information is linearly decodable at the channel's address at all (AUC 0.504 vs. Gemma's 0.807). Instead, Qwen's outcome information is weakly decodable at the *final prompt token* (0.66–0.81 across two independent 96-row draws), and writing the class-mean at that readable site does not repair (label-shuffled matches real exactly, paired $+0.005$; a sixth confirmed prediction). The anatomy is conserved, its coordinates are not: **Gemma separates gauge from lever by subspace at one position; Qwen separates them by position at one layer. Gemma concentrates, Qwen distributes — the recipe is cross-model, the compression is not.**

6 Discussion and Limitations

Location is not control, in either direction. The program's two dissociations bound what probe evidence can license. For monitoring: a high-AUC correctness probe can be a perfectly good gauge while reading a subspace the computation never consults — deletable at zero cost. For steering: the potent content can live where no probe finds it, so steering nulls on readable directions are uninformative about controllability. The lineage's raw difference-in-means vectors plausibly pay a dilution tax and a damage ceiling that causal filtering removes — a testable prediction (in Gemma, projection was necessary; in Qwen, where the state distributes, raw suffices and projection dilutes), which the concentrated-vs-distributed contrast turns into a measurable, model-level design property.

Endogeneity, honestly. The channel repairs without being decodably privileged, so we claim *exogenous mediation*: the vector carries content *about* natural success without being shown to be the model's own per-row decision variable (as far as linear tools see). The removal experiments sharpen this: at the lever's site, natural success requires the state's specific content wholesale — replacing it with a typical state at preserved energy abolishes success, as does keeping only the 8 repair dimensions — so the channel is a write-compact port into a broadly-carried state, not a thin wire that computation runs through. We say "an identified sufficient channel," not the mechanism; sufficiency was searched for, and pre-registration is the mitigation we offer, not a proof of uniqueness.

The gauge is not the verbal channel — and not useless. Asked to judge its own replayed answers (a 2,000-row census), the model verbalizes "no" near-uniformly: on 85% of answers it in fact got right, with 97.5% unanimity across samples and a confident-wrong cell of 0.9% — while the linear gauge reads upcoming correctness at AUC 0.94 on the same task. Whatever the probe reads, it is not the model's verbal self-report channel. And the gauge, causally inert as a lever, turns out to do causal *work* in the one role its nature permits: as a *selector*, ranking steered candidate futures well enough to pick the gold write-address (Section 5.5). Gauge and lever remain different objects; the gauge's job is reading — and it reads steered states as well as natural ones.

Open anomalies and limits. Misdirection asymmetry remains open: repair transfers, misdirection does not, and the attractor-strength account we tested has the wrong sign. Positional addressing is the one answer-adjacent ingredient of the answer-free repair (every pre-named answer-free position policy fails). Quantitative claims rest on n=16–46 row sets with MDEs of 0.10–0.13 on nulls; positive results are one task family, one model (with the stated Qwen and subtype scope); brittleness and entropy telemetry are unrun. Future work, in registered order: the layer$\times$position information map in Qwen; closing the repair loop by composing the gauge-selector with the transferring write protocol; and the site-specificity control for the necessity coda (is any position's content this necessary, or only the lever's site?).

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A Pre-registration excerpts and row provenance

[TODO: Appendix — artifact-checklist table, parse/baseline diagnostics, negative-results catalog, provenance figure, registration excerpts.]